

day-77-paperclip-bridge-integration

Day 77 - Paperclip + Bridge Integration

Status: PASS for developer-side Bridge gating

Blocker class: NONE for the gated task request path

Service blocker still observed: paperclip_sandbox_service_not_running

Lane

Paperclip protected actions through Bridge approval.

What Was Implemented

- Wired 'POST /api/bridge/paperclip/tasks' to create a Bridge approval request for the scoped Paperclip task-create action.
- Kept Paperclip writes and execution disabled.
- Kept the request in pending owner approval state instead of dispatching it.
- Added route tests proving the approval request payload is scoped and does not include the raw request body.

Files Changed

- 'src/app/api/bridge/paperclip/tasks/route.ts'
- 'src/lib/paperclip-bridge-routes.test.ts'
- 'runtime/day-77-paperclip-bridge-integration-proof.json'
- 'runtime/day-77-paperclip-bridge-integration.md'
- 'runtime/day-77-paperclip-bridge-integration.pdf'

Routes Changed

- 'POST /api/bridge/paperclip/tasks'

UI Behavior

No Mission Control or Gateway designer HTML, CSS, class names, or visual assets were changed.

This lane only changes the backend truth for Paperclip protected task creation.

The owner-facing Paperclip state can now truthfully show:

- owner approval required
- Bridge Session required
- approval pending after request
- execution disabled
- writes disabled
- request not dispatched

Service Runtime Behavior

Runtime proof was performed against a short-lived local standalone Mission Control runtime on '127.0.0.1:3337'.

The proof used an in-process API key and did not write or modify any '.env' file.

Proof result:

- 'GET /api/bridge/paperclip/status' returned HTTP 200 with blocker 'paperclip_sandbox_service_not_running'.
- 'POST /api/bridge/paperclip/tasks' returned HTTP 409 because protected execution still requires approval/session.
- The POST response created a fresh pending Bridge approval request.
- The POST response included an audit event id.
- The POST response kept 'accepted_for_execution:false'.
- The POST response kept 'request_dispatched:false'.
- The POST response kept 'execution_enabled:false'.
- The POST response kept 'writes_enabled:false'.

- The POST response kept 'protected_actions_enabled:false'.

Paperclip sandbox service recovery remains a separate service lane, but the Bridge gating path for Paperclip task creation is proven.

Proof artifact:

- 'runtime/day-77-paperclip-bridge-integration-proof.json'

Tests Run

- 'git diff --check'
- 'pnpm run typecheck'
- 'pnpm run build'
- 'pnpm exec vitest run src/lib/paperclip-bridge-routes.test.ts src/lib/paperclip-bridge.test.ts'
- 'pnpm test'
- 'node scripts/protected-route-smoke-contract.mjs http://127.0.0.1:3337'
- 'node scripts/check-protected-file-invariants.mjs'
- 'node scripts/secret-scan-contract.mjs'
- 'node scripts/raw-exposure-scan-contract.mjs'
- '.env' diff check

Results:

- Typecheck: PASS
- Build: PASS
- Focused Paperclip tests: 42 passed
- Full test suite: 198 files passed, 1480 tests passed
- Protected route smoke: PASS, 41 routes checked
- Protected-file invariant scan: PASS
- Secret scan: PASS
- Raw exposure scan: PASS
- '.env' diff: clean

Deploy / Restart / Smoke Result

- No public exposure was added.
- No persistent production restart is claimed in this lane.
- A local standalone proof runtime was started on '127.0.0.1:3337', used for the gated Paperclip task proof and route smoke, then stopped.
- No listener remained on port 3337 after proof.

Remaining Blocker

Paperclip sandbox service is still not running:

- blocker: 'paperclip_sandbox_service_not_running'
- blocker classification for Paperclip sandbox service: SERVICE_DOWN

This does not block Day 77 developer-side Bridge gating because the protected Paperclip task action is now approval-gated and does not execute or write.

Rollback

After commit:

```
""bash
git revert <day-77-paperclip-bridge-commit>
""
```

Commit / Push

Pending at report generation time.

Next Day

Day 78 starts next: SpaceAgent + Bridge integration.